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Debugging TCP/IP Communications

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Why Not Use a Source Level Debugger?

- Real-time nature of most protocols prevents stopping execution for long periods of time
- Source level debugger might work for direct serial connection with simple protocols
- Serial port is generally in use when TCP/IP is active

General Tips

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- Test your application using POSE first
- Instrument your app for on-screen or post-mortem analysis

Alternate Debugging Methods

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- Printf
- Audio cues
- Writing to the screen
- Asserts
- Hardware
- Logging

Printfs

- DbgMessage sends text messages to the debug console
  - Won’t work over serial debug link
  - Slow when used with Poser
- Use HostFPrintF when debugging using Poser

Audio Cues (aka ‘beep’)

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- SndPlaySystemSound(sndError)
- Usually not a great way to debug

Writing to the Screen

- Have an area on the display reserved for status / error info
- Can get busy if too many of these status areas are used
- Lose context if a lot of information is being overwritten to the same area

"asserts"

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- ErrNonFatalDisplay
- Good way to catch unexpected situations

Hardware

- Second serial port for debugging
  - PalmPilot upgrade card has 2nd serial port, so serial port can be used for debugging if application uses IR
  - Think of POSE as a second serial port
- Serial Analyzer/Protocol Analyser
  - Nice if you have one
- Use a second host running TcpDump or EtherPeek or equivalent

Logging

- Use conditional macros so logging info can be selectively chosen based on problem
- Log to circular RAM buffer
  - Otherwise writing to database interferes with serial communication timing
- Write log to memo pad entry(s) at close of serial device
- Alternatively, write log to HotSync log

Logging

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- Some examples…

PalmDebugger w/ Real Device

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- Connect to debugger before testing
  - Use shortcut . 1
  - Prevents hangs in ErrorDisplay dialog
- Connect to debugger after Debugger is called (or crash occurs)
  - May have to change connection speeds to match serial device speed to successfully attach to debugger

Poser—NetLib emulation

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- Poser redirects to host TCP/IP stack
- Great for troubleshooting high level issues
- Can mask low level timing or memory issues

PalmDebugger w/Poser

- Doesn’t use serial to communicate with Poser, so doesn’t get in way of serial driver
- Can test (and debug fatal situations) without the need for a real device

Poser—HostControl

- Better way to do "Printf"
- Another way to do logging
- May not be appropriate for all situations
  - Some things require real device (Ir, Palm VII)

Header Files for Info Mentioned

- (All in Incs/Core/System)
- DLServer.h
- ErrorMgr.h
- HostControl.h


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